AMC 12 · November Full Mock

Worked solutions • 25 questions • 75 minutes

Question 1 · Answer B

A mixture uses syrup and water in ratio 2:7 by volume. After 6 litres of syrup are added to 27 litres of mixture, what fraction of the new mixture is syrup?

Choices: A 2/5 B 4/11 C 2/9 D 5/11 E 1/3

Initially there are $27\times 2/9=6$ litres of syrup. The new mixture contains 12 in a total of 33 litres, hence 4/11.

Answer: B · 4/11

Question 2 · Answer B

A staircase of display cubes uses rows of 2, 5, 8, … cubes. How many cubes are in its first 12 rows together?

Choices: A 228 B 222 C 204 D 210 E 216

The last row has $2+11\times 3=35$. Pair first and last; there are six pairs each totalling 37, so 222 cubes.

Answer: B · 222

Question 3 · Answer C

A number x satisfies log base 3 of x minus log base 3 of 4 equals 2. What is x?

Choices: A 27 B 12 C 36 D 18 E 81

Combine the logarithms to get $\log_3(x/4)=2$. Thus $x/4=9$ and $x=36$.

Answer: C · 36

Question 4 · Answer C

A committee of 12 has mean age 30. Two members with mean age 42 leave. What is the mean age of the remaining members?

Choices: A 27 B 26 C 138/5 D 29 E 28

The total age decreases from 360 to 276. Dividing by 10 remaining members gives 138/5.

Answer: C · 138/5

Question 5 · Answer A

What is the last digit of $3^{2026} + 7^{2026}$?

Choices: A 8 B 4 C 2 D 6 E 0

Both last-digit cycles have length four. Since $2026 \equiv 2 \bmod 4$, both terms end in 9; their sum ends in 8.

Answer: A · 8

Question 6 · Answer C

In a regular polygon, an interior angle is seven times its exterior angle. How many diagonals does the polygon have?

Choices: A 120 B 112 C 104 D 88 E 96

Eight exterior angles equal 180°, so one is 22.5°. There are 16 sides. Remove the 16 sides from $C(16,2)=120$ pairs of vertices to get 104 diagonals.

Answer: C · 104

Question 7 · Answer C

Let $z=(3+i)/(1-i)$. What is $|z|^2$?

Choices: A 5/2 B 2 C 5 D 10 E 8

Squared moduli multiply and divide: $|3+i|^2/|1-i|^2=10/2=5$.

Answer: C · 5

Question 8 · Answer E

A positive integer is called a split-total if it is represented by the ordered sum of three positive odd numbers. How many representations does 15 have?

Choices: A 21 B 35 C 36 D 15 E 28

Write each odd part as $2u+1$. Then $u+v+w=6$ in nonnegative integers. Stars and bars gives $C(8,2)=28$.

Answer: E · 28

Question 9 · Answer C

An acute angle θ has $\tan\theta=3/4$. What is $\cos(2\theta)$?

Choices: A 4/5 B 24/25 C 7/25 D 12/25 E 3/5

Use a 3–4–5 right triangle: cosine 4/5 and sine 3/5. Their squared difference is $(16-9)/25=7/25$.

Answer: C · 7/25

Question 10 · Answer D

Find the exponent of 3 in 36! divided by 12!.

Choices: A 8 B 13 C 17 D 12 E 10

Legendre's count gives 12+4+1=17 factors of 3 in 36! and 4+1=5 in 12!. The quotient contains 12.

Answer: D · 12

Question 11 · Answer D

A sequence has $b_0=4$ and $b_{n+1}=3b_n-2$. What is b_5?

Choices: A 486 B 729 C 972 D 730 E 243

Subtract the fixed point 1: $b_{n+1}-1=3(b_n-1)$. Therefore $b_5=1+3\times3^5=730$.

Answer: D · 730

Question 12 · Answer B

Four fair six-sided dice are rolled. Let X be the number of different face values displayed. What is E[X]?

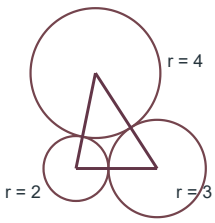
Choices: A 7/2 B 671/216 C 4 D 65/24 E 3

For one face the probability of appearing is $1-(5/6)^4$. Sum the six face indicators: $6[1-625/1296]=671/216$.

Answer: B · 671/216

Question 13 · Answer A

Three mutually tangent circles of radii 2, 3 and 4 are externally tangent in pairs. What is the area squared of the triangle joining their centres?



Choices: A 216 B 196 C 108 D 144 E 225

The three centre distances are 5, 6 and 7. The semiperimeter is 9. Heron gives $\text{area}^2=9\times4\times3\times2=216$.

Answer: A · 216

Question 14 · Answer A

The positive coefficient c is chosen so x^3-4x^2+cx-6 has root 2. What is the sum of the squares of the other two roots, counted with multiplicity?

Choices: A -2 B 0 C 2 D 4 E 1

Substitution at 2 gives $8-16+2c-6=0$, so $c=7$. Division by $x-2$ leaves x^2-2x+3. Its roots have sum 2 and product 3, so the sum of their squares is $4-6=-2$.

Answer: A · -2

Question 15 · Answer A

How many distinct solutions in $0 \le x < 2\pi$ satisfy $\sin(3x)=\sin(x)$?

Choices: A 6 B 5 C 4 D 7 E 8

Equal sines give $3x=x+2k\pi$ or $3x=\pi-x+2k\pi$. The first gives two multiples of π; the second gives four odd multiples of $\pi/4$. The lists are disjoint, for 6.

Answer: A · 6

Question 16 · Answer E

Eight different letters include A, B and C. In how many permutations do A and B occur next to each other while B and C do not?

Choices: A 7200 B 11520 C 7920 D 10080 E 8640

Contract AB or BA: $2\times7!$ permutations. With both adjacencies, B is in the middle of ABC or CBA, giving $2\times6!$. Subtract to get 8640.

Answer: E · 8640

Question 17 · Answer B

A positive integer in base b is written 121; in base b+2 it is written 81. What is its decimal value?

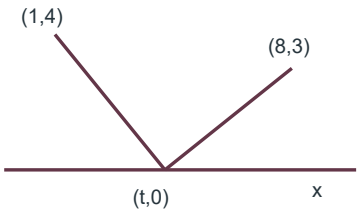
Choices: A 121 B 81 C 144 D 100 E 64

The two expressions are $(b+1)^2$ and $8b+17$. Equating gives $(b-8)(b+2)=0$. Only $b=8$ is a valid base; 121 in base 8 is $64+16+1=81$.

Answer: B · 81

Question 18 · Answer E

A point (t,0) on the x-axis is joined by line segments to (1,4) and (8,3). What is the square of the least possible total length?



Choices: A 74 B 81 C 100 D 50 E 98

Reflect (1,4) to (1,−4). The two segments then form a broken line no shorter than the straight line to (8,3). That segment crosses the axis, so equality occurs. Its squared length is $7^2+7^2=98$.

Answer: E · 98

Question 19 · Answer B

The polynomial x^4+ax^2+16 has four distinct real roots that are all integers. What is the sum of all possible values of $|a|$?

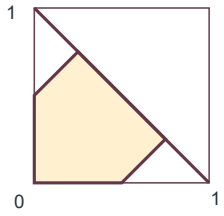
Choices: A 8 B 17 C 20 D 16 E 25

Roots occur as $\pm u, \pm v$ with distinct positive u,v. Their squares multiply to 16, so uv=4 and the only distinct pair is 1,4. Thus $a=-(1+16)=-17$, giving 17.

Answer: B · 17

Question 20 · Answer B

A point is uniformly distributed in the unit square. What is the probability that its distances to the two coordinate axes have sum less than 1 and differ by less than 1/2?



Choices: A 1/4 B 3/8 C 7/16 D 5/16 E 1/2

The triangle $x+y<1$ has area 1/2. The line $x-y=1/2$ cuts off a triangle with base 1/2 on the x-axis and height 1/4, of area 1/16. The corresponding y-side triangle is congruent. Subtract twice 1/16 to obtain 3/8.

Answer: B · 3/8

Question 21 · Answer E

Define $f(x)=(x-1)/(x+1)$. Starting with $x=3$, apply f exactly 2026 times. What is the absolute value of the result?

Choices: A 2 B 2/3 C 3 D 1/2 E 1/3

The first four values after starting are $1/2$, $-1/3$, -2, 3. They form a cycle of length four. Since $2026 \equiv 2 \bmod 4$, the result is $-1/3$, whose absolute value is $1/3$.

Answer: E · 1/3

Question 22 · Answer A

How many different circular necklaces have eight beads, exactly four black and four white, if rotations are identified but mirror images are not identified unless a rotation matches them?

Choices: A 10 B 8 C 12 D 9 E 14

Average the fixed colourings over the eight rotations. Identity fixes 70; the half-turn fixes $C(4,2)=6$. Each quarter-turn fixes 2; odd turns fix none. Thus $(70+6+2+2)/8=10$.

Answer: A · 10

Question 23 · Answer D

What is the sum of $\gcd(k,30)$ over $1 \le k \le 30$?

Choices: A 150 B 180 C 105 D 135 E 120

For each divisor d of 30 the count of k with $\gcd=d$ is $\varphi(30/d)$. For $d=1,2,3,5,6,10,15,30$ the contributions are $8,16,12,10,24,20,15,30$ respectively. Their sum is 135.

Answer: D · 135

Question 24 · Answer D

Let ω be a nonreal cube root of unity. Compute the squared modulus of $(1-\omega)^5/(1-\omega^2)^2$.

Choices: A 81 B 18 C 3 D 27 E 9

The two nonreal cube roots are conjugates. Each squared distance to 1 is $(1+1/2)^2+3/4=3$. The squared modulus of the quotient is $3^5/3^2=27$.

Answer: D · 27

Question 25 · Answer E

Select a subset of {1,2,…,30} in which no chosen number is twice another chosen number. What is the largest possible size?

Choices: A 18 B 19 C 15 D 21 E 20

Partition by odd part d into doubling chains d,2d,4d,… . On a chain of length L, at most ⌈L/2⌉ can be chosen, attained by alternating entries. The chain lengths for odd d=1,3,…,29 are 5,4,3,3,2,2,2,2,1,1,1,1,1,1,1. Their alternating maxima sum to 20.

Answer: E · 20
